

Pranav Raj

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EDUCATION

Manipal University Jaipur

B.Tech in Information Technology

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Cloud Computing, Machine Learning

2022 – 2026

CGPA: 7.8/10

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript, TypeScript
- **Backend & Payments:** FastAPI, Node.js, REST APIs, WebSockets, Idempotency, Circuit Breaker, Rate Limiting, HMAC Webhooks
- **Frontend:** React, Vite, TailwindCSS, Chart.js, HTML5 Canvas
- **Databases:** PostgreSQL, MySQL, Redis, SQL Query Optimization
- **Cloud & DevOps:** AWS, Docker, Terraform, Git, GitHub Actions, CI/CD, Linux
- **Core & Testing:** Data Structures & Algorithms, OOP, System Design, Pytest, Vitest
- **AI/ML:** PyTorch, TensorFlow, TensorFlow.js, Scikit-learn, LangChain, FAISS, HuggingFace, RAG

ACHIEVEMENTS

- **Codeforces Specialist** (Rating: 1459) | **LeetCode** (Rating: 1415, 300+ solved) | **GeeksforGeeks** (200+ solved, Institute Rank: 113)
- **Deloitte Hackathon Finalist** for an AI-powered invoice fraud detection system
- Research paper accepted at **Springer LNNS (ICT4SD 2026)**; won **Best Major Project** for the same work

INTERNSHIP

Backend Analytics Intern, NIT Mizoram

Certificate Jun 2023 – Aug 2023

- Developed **5+ REST APIs** in Python and FastAPI for backend analytics and data processing
- Optimized **SQL queries and data pipelines**, reducing dashboard response latency by **15%**
- Debugged backend issues using logs, improved workflow reliability, and wrote technical documentation

PROJECTS

NIYAM – Policy Enforcement Gateway for AI-Agent Payments

GitHub | **Razorpay AI Buildathon**

- Built a **FastAPI** gateway between AI agents and Razorpay (test mode): an LLM compiles spending rules into a **JSON policy** once, then payments are gated by **deterministic code** with no LLM in the payment path (<0.04ms)
- Implemented **idempotency keys**, a **circuit breaker** that fails closed (503) on rail outages, **token-bucket rate limiting**, **HMAC-SHA256 webhook verification**, and hashed API keys
- Added an **append-only audit ledger** with rule-fired explanations, CSV export, and a signed-token human approval flow; covered by **27 Pytest cases**

Blast Radius – Consequence-Aware AI Governance Plane

GitHub | **Accenture Innovation Challenge**

- Built a proxy that scores each AI action on **irreversibility, people affected, regulated data, and financial value**, routing it to 4 risk tiers so the LLM safety judge runs only on high-risk requests
- Implemented gates for **PII redaction, prompt-injection detection, and commitment/claim checks**, plus mandatory human review for critical actions such as large wire transfers
- Issued **SHA-256 signed audit receipts** per decision; built with **React + TypeScript** and tested with **Vitest** across 14 banking, lending, and healthcare scenarios

Distributed Job Orchestrator (Atlas)

GitHub

- Designed a job orchestration system with **FastAPI, Redis, and PostgreSQL** to accept, prioritize, and run jobs across concurrent workers
- Implemented **202 Accepted non-blocking ingestion**, priority queues, automatic retries with exponential backoff, and graceful failure handling
- Added **Pytest** coverage, Alembic async migrations, Docker containerization, and OpenAPI/Swagger documentation

AI Invoice Fraud Detection & Automation

Kaggle | **Deloitte Hackathon**

- Built an **OCR-driven document pipeline** for automated invoice verification and fraud detection
- Applied transfer learning (**MobileNetV2, DenseNet121, EfficientNet**) for invoice classification
- Achieved **84%+ accuracy** and ranked in the **Top 8 teams in 300**

CERTIFICATIONS

- **ML and Deep Learning Specializations** (Andrew Ng) | **Python Essentials 1 & 2** (Cisco)